ENTERPRISE EDITION 2026 • EXECUTIVE PRODUCT CATALOG & PROPOSAL

SMART BLOOD BANK & DONOR MANAGEMENT PORTAL

A mission-critical healthcare platform for blood banks, thalassemia welfare societies, and humanitarian trusts. Built to automate donor matching, eliminate operational bottlenecks, and save lives in real-time.

12 Modules	Live QR Code	1-Click WhatsApp	English + اردو
FULL LIFECYCLE SUITE	PUBLIC INSTANT VERIFICATION	ZERO GATEWAY COSTS	NATIVE NASTALIQ RTL

Target Audience: NGO Board of Trustees, Healthcare Directors, Medical Camp Coordinators • Confirmed Production Ready

The Operational Challenges in Blood Banking Today

1. Fatal Delays in Matching

Coordinators search through thick paper registers or disorganized spreadsheets. In critical ICU emergencies, manual search takes 45–60 minutes, risking patient survival during surgeries and trauma.

2. Unverified Donor Claims

Hospitals have no way to verify if a donor is genuine, safe, or on medical cooldown. Unscrupulous blood agents exploit paper cards, jeopardizing patient health and blood purity.

3. Heavy SMS & Paper Costs

Traditional SMS gateways cost Rs. 25,000–45,000 monthly, with low open rates. Paper ledgers tear, lose legibility, and cannot provide compliance reports for foreign grant audits.

✔ How Our Portal Completely Solves This Crisis:

- **Sub-3-Second Matching:** Multi-city search instantly filters eligible donors by group and cooldown.
- **1-Click Free WhatsApp:** Pre-filled emergency request dispatch via `wa.me` protocol with zero SMS costs.
- **Instant QR Verification:** Doctors scan the donor's digital card to verify live eligibility and donation history.
- **Automated 90-Day Engine:** Automatic pulse indicator prevents premature donations per WHO standards.

Traditional Paper System vs Smart Digital Portal

OPERATIONAL FUNCTION	TRADITIONAL MANUAL METHOD	SMART DIGITAL PLATFORM	IMPACT / GAIN
Emergency Matching	Leafing through physical notebooks (45–60 mins)	Smart algorithm matches eligible donors (< 3 seconds)	95% FASTER
Donor Messaging	Typing manual SMS or paying third-party gateways	1-Click WhatsApp Direct Emergency Dispatch (Pre-filled)	100% FREE
Donor Verification	Unverified paper slips or photocopies	Live Dynamic QR Code scannable with any smartphone	ZERO FRAUD
Public Access	Family must visit center or dial busy landlines	24/7 Public Request Portal with unique tracking code	24/7 ONLINE
Eligibility Protocol	Manual date calculation errors lead to premature calls	Automated 90-Day Medical Cooldown countdown pulse	100% SAFE
Donor Retention	No recognition; donors donate once and disappear	Bronze-to-Platinum Gamification & PDF Certificates	3X REPEATS
Executive Reporting	3–4 days of manual ledger tallying	Instant exports in Excel (.xlsx), PDF, and Word (.docx)	INSTANT

< 3 Sec
MATCHING RESPONSE

Rs. 0
MESSAGING TOLLS

100%
PAPERLESS AUDIT

300%
REPEAT DONOR BOOST

Donor Intelligence, 90-Day Cooldown & Bulk CSV Migration

Automated 90-Day Medical Cooldown Engine

In accordance with WHO and Red Crescent safety standards, the system calculates exact remaining cooldown days from the last donation. Displays a green **Eligible Now** pulse or amber countdown (e.g. `24d remaining`).

Excel / CSV Bulk Migration Tool

Upload thousands of historical donor records from CSV spreadsheets in seconds. Features sample CSV template download, automatic duplicate detection, and automated city-to-ID mapping.

Real-Time Duplicate Prevention

Cross-references CNIC and phone numbers dynamically during online public sign-ups and admin manual entry. Eliminates duplicate profiles and ensures consolidated medical histories.

Filtered CSV Data Export

Instantly export filtered donor rosters by blood group, city, or status directly into Excel-ready CSV format for mobile camp planning, auditing, and offline emergency calls.

Automated Standardized Registration Code

Generates official unique hospital codes based on City, Initials, and Phone (e.g.,`LHR-MA-12345`), ideal for indexing on blood bags, lab reports, and physical PVC cards.

Student & University Ambassador Network

Flags student donors and tracks university campus affiliations, empowering the NGO to organize targeted campus blood camps during critical blood shortages.

Emergency Matching & Free 1-Click WhatsApp Dispatch

1. **Patient Request**
   Blood group, hospital, units & urgency level logged.
2. **Smart Matching**
   System instantly filters eligible, un-cooldown donors.
3. **1-Click WhatsApp**
   Pre-formatted emergency dispatch text sent instantly.
4. **Call Logging**
   Coordinator logs outcome (Agreed / Donated).

100% Free WhatsApp Direct Messaging

Uses the official `wa.me` protocol to open chats directly with verified donors. No third-party API subscription, no monthly character fees, and no risk of gateway downtime.

Multi-Parameter Proximity Matching

Ranks donors based on blood compatibility, geographic district/city, and historical responsiveness, ensuring the closest available hero is notified first.

Bilingual Emergency Message Templates

Pre-populates patient name, hospital, units needed, urgency, and direct helpline in Urdu or English, saving precious minutes during critical surgical emergencies.

Coordinator Performance CRM

Records outbound call duration, agent name, and donor response, enabling management to monitor response times and staff fulfillment rates.

Digital Donor Cards & Live Public QR Verification Portal

Official Dual-Sided Printable PVC Cards

Renders official donor ID cards complete with NGO logo, donor photograph, blood group, registration number, emergency contacts, and personalized dynamic QR code.

Batch Card Print Queue

Automated printing queue flags new donors who haven't received physical cards yet. Coordinators select multiple donors and print full sheets for thermal PVC card printers in seconds.

Permanent Fraud Protection

Card design contains cryptographic verification tokens, making it impossible for unauthorized persons to forge donor cards or counterfeit safe blood claims.

Live Public Verification Portal (/portal/verify-donor)

Scanning the card's QR code with any smartphone camera opens a dedicated public verification page without requiring any app installation or staff login.

Real-Time Medical Status on Scan

Doctors and hospitals see live data:

- • **Eligibility Pulse:** Verified Eligible Now or Days remaining.
- • **Honor Tier Badge:** Platinum Hero / Gold Lifesaver.
- • **Donation Milestones:** Verified count of lifetime donations.
- • **Reliability Score:** Track record of honored emergency calls.

Public Emergency Portal & Blood Inventory Control

24/7 Public Request Portal (/portal/request-blood)

Enables patient families and hospital coordinators to submit urgent blood needs online 24/7 without creating an account. Captures patient name, required units, hospital, doctor, and contact info.

Live 8-Blood Group Inventory Control

Real-time tracking of available, collected, and reserved blood bags across all major groups (A+, A-, B+, B-, AB+, AB-, O+, O-).

Instant Request Tracking Code (REQ-00042)

Upon submission, patients receive an official tracking reference and a direct coordinator helpline button, keeping anxious families informed and reassured in real time.

Critical Shortage & Expiry Alerts

Color-coded threshold alerts warn coordinators when rare blood groups (e.g. O-, AB-) drop below safety stock, triggering early camp announcements before stocks hit zero.

Anti-Spam & Rate-Limiting Protection

Throttles submissions per IP address and sanitizes all inputs to protect the NGO from automated web spam while keeping emergency channels open.

Public Transparency Page (/portal/availability)

Public inventory transparency page allows doctors and patients to check blood group availability in real time before traveling to the center.

Gamification Badges, Official PDF Certificates & Campaigns

5-Tier Honor Badge Gamification System

Rewards repeated life-saving milestones, dramatically increasing donor retention:

- **PLATINUM HERO** (10+ Donations)
- **GOLD LIFESAVER** (5–9 Donations)
- **SILVER CHAMPION** (3–4 Donations)
- **BRONZE SAVER** (1–2 Donations)

Official Printable PDF Certificates

Generates official branded appreciation certificates featuring donor's tier, lifetime donations, NGO watermark, authorized signatures, and embedded verification QR code.

Financial Donations & Automated Receipts

Tracks cash, online, bank transfer, and cheque monetary donations for NGO medical operations. Generates printable official receipts with tax registration numbers and donor records.

Campaign & Mobile Drive Attendance Tracker

Coordinates university, college, and corporate blood camps. Generates printable attendance registers and logs volunteer staff hours and donor turnouts.

Public Story & Hero Showcase

Admin tool to publish real-life patient survival stories and donor testimonials onto the public landing page to build community trust and drive new registrations.

Bilingual Urdu RTL / English, Dark Mode & Audit Security

1-Click Bilingual Urdu (اردو) & English Switcher

Animated iOS-style pill switcher in the top navigation toggles the entire dashboard between English and native Urdu seamlessly, empowering non-English speaking field staff.

Triple-Format Report Exporter (Excel, PDF, Word)

Export comprehensive reports for donors, donations, financials, and requests in Microsoft Excel (.xlsx), Adobe PDF (.pdf), and Microsoft Word (.docx).

Authentic Noto Nastaliq Typography & True RTL

Complete Right-to-Left (RTL) layout with beautiful Google Noto Nastaliq Urdu typography. Mirrors form inputs, data tables, and navigation icons cleanly.

Role-Based Access Control (RBAC)

Powered by Spatie. Granular permissions separate Super Admins, Branch Managers, Calling Coordinators, and Lab Technicians, securing sensitive donor data.

Day / Night Dark Mode Switch

Persistent sun/moon dark theme switch with zero-flash startup. Protects coordinator eyesight during late-night emergency hospital shifts.

Immutable Audit Trail & Forensic Logging

Every sensitive database edit, card reprint, and deletion is recorded permanently with timestamp, user ID, and IP address for complete transparency.

Licensing Packages & Implementation Roadmap

STARTER TRUST EDITION

Rs. 150,000

Small Welfare Trusts & City Societies

- Single Branch Deployment
- Core Donor Registry & Cooldown
- Smart Emergency Matching
- 1-Click Free WhatsApp Messaging
- Excel & PDF Report Exports
- 3 Staff Accounts • 30-Day Support

RECOMMENDED

PROFESSIONAL CENTER EDITION

Rs. 250,000

Thalassemia Centers & Hospitals

- **All Starter Features +**
- Digital QR Verified Donor Cards
- Public Emergency Request Portal
- Bulk CSV Migration & Export Tool
- Gamification Tiers & PDF Certificates
- Full Urdu RTL + English Localization
- Persistent Day/Night Dark Mode
- Unlimited Users • 90-Day Priority SLA

ENTERPRISE NETWORK

Rs. 450,000+

Nationwide Trusts & Red Crescent

- **All Professional Features +**
- Multi-Branch City Network
- Dedicated Cloud Server & SSL
- Custom Hospital ERP Integration
- White-Label Branding & Domain
- Automated Cloud Backup Schedule
- 1 Year AMC • 24/7 Dedicated Support

Annual Maintenance Contract (AMC): Ongoing security patching, database backups, and server monitoring available at **Rs. 10,000 – 15,000 / month**.